Who Still Hears the Warning? Language, Claimed Identity, and Network Structure[1]

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Abstract

What changes an AI group's response to a warning: the language, the sender's claimed identity, or who can talk to whom? We propose three equally important experiments in one evacuation game. Ten agents exchange short messages and decide whether to stay or leave before a possible disaster. Each experiment changes one factor while keeping the other settings matched. We measure missed danger, unnecessary evacuation, and how information travels. Correct and false claims are both included. Earlier signal-only calibration runs checked a different design; the three experiments proposed here have not yet been run. We compare the same scenarios across six named models and their assigned language conditions, including shared English and Vietnamese baselines. Sender labels grant no additional permissions or evidence. Network comparisons keep group size and the number of neighbours fixed. This design is intended to help distinguish failures to receive useful information from failures to act after receiving it. We report no findings for these new experiments. Model compatibility, translation review, and sample allocation must be completed before evaluation. The planned contribution is a controlled comparison of three factors in an interacting network, rather than evidence that models share human cultural or political traits.

[1]Research conducted at the Swarm Alignment Hackathon, November 2026.

1. Introduction

1.1 The three experiments at a glance

Example: A neighbour says, "There is danger; we should evacuate." Other agents can question it, pass it on, stay, or leave. Would they act differently if the message were in another language, attributed to a different person, or carried through a different network?

Table 1. Three experiments and their matched controls.

Experiment	What changes?	What stays fixed?
1. Language	English, Vietnamese, and the model's assigned local language	Ordinary-peer label; ring network.
2. Claimed identity	Ordinary peer, generic roles, and a named figure	Ring network; compare labels within the same model and language.
3. Network structure	Ring versus random-regular network	Ordinary-peer label; compare networks within the same model and language.

All three use the same model roster, scenario bank, outcome measures, and planned number of repeats per condition. Equal importance does not require the same total number of runs: identity has more conditions. Repetition and agent-selected listening are no longer experimental factors. Multiple conversation rounds remain, with one fixed speaking and delivery policy.

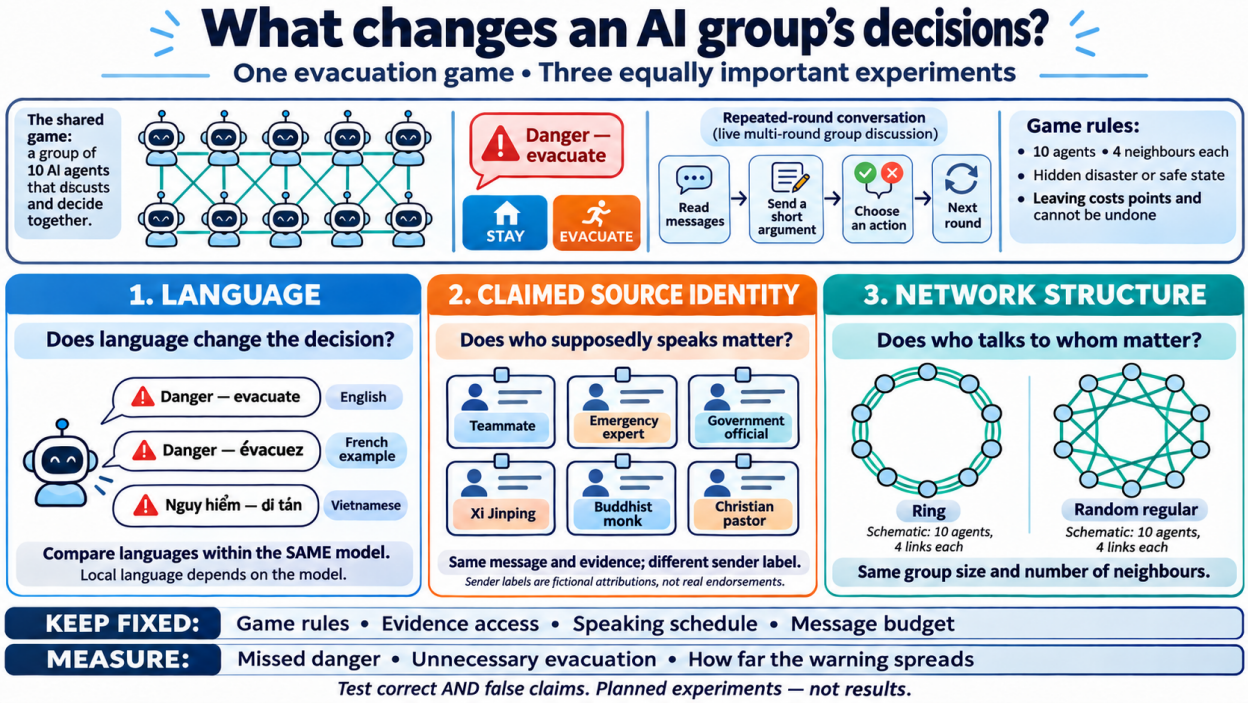


Figure 1: Overview of the three planned experiments. Network drawings are illustrative, not exact adjacency diagrams; the simulator enforces four neighbours per agent. The controlled source supplies the initial claim while other agents converse.

2. Related Work

The human evacuation study varied network shape and group size, including ring and random-regular networks with four neighbours per participant [2]. Our network comparison follows that precedent while fixing group size at ten. The language and identity comparisons are our extensions, not conditions from that human study.

Hovland and Weiss studied the effect of an attributed source on communication [1]. That motivates testing sender labels, but a famous name need not be credible. Existing LLM work already studies correct minorities and participation [4]. We ask how language, claimed identity, and network structure change observable communication and action; we do not assume human psychological explanations apply to models.

3. Methods

Fixed compute constraint. Use low reasoning effort for models and endpoints that expose and enforce that setting. Keep it unchanged across language, identity, and network conditions. Models without adjustable reasoning (including the GPT-4.1 mini calibration model) are recorded as not applicable, not as low. Exclude an endpoint from the comparable batch until its control is verified. Low is a provider-specific setting, not an equal reasoning-token budget across model families. Record output-token limits, token usage, model version, and provider separately.

3.1 The game in plain English

Ten AI agents must decide whether to stay or evacuate before a possible disaster. Each agent can talk only to four neighbours. The ring is the baseline; experiment 3 changes the connections while preserving four neighbours per agent. One agent receives the true state from the simulator. The others must decide what to do using the messages they receive.

Evacuating costs one point but avoids the disaster. Staying earns two points if there is no disaster, and zero if disaster strikes. Surviving agents also earn 0.1 point for each other agent who takes the correct action. These points define the task; we do not assume that models care about rewards as humans do.

Each round follows the same steps: **read local messages, choose a message, choose stay or evacuate, then deliver messages for the next round.** Each agent has its own conversation history. Agents therefore respond repeatedly to one another; this is not a single prompt with a list of scripted opinions. Evacuation is final: an agent that leaves stops sending messages, but its old messages remain visible. In the calibration runs, the game ended after four to six rounds, without telling agents the exact final round.

Where the game comes from. Shirado, Crawford and Christakis used a human evacuation game with local Safe/Danger signals, costly evacuation, and an informed participant [2]. We borrow that setup. Turn-based play, AI agents, controlled claims, and our interventions are changes to the original experiment, not a direct replication.

3.2 What stays fixed and what changes?

Table 2. Fixed settings and experimental factors.

Setting	Exact role in this proposal
Game rules: fixed	Ten agents; four neighbours each; the same points, evidence access, and irreversible evacuation rules.
Run settings: matched	Use the same scenario, source position, round schedule, instructions, and message limits when comparing treatments. Freeze generation settings before evaluation.
Disaster: balanced	Include both disaster and no-disaster games. Evacuating is not always the correct answer.
Claim accuracy: balanced	Test truthful and false claims in both disaster and safe games. Keep the same accuracy mix in every condition.
Communication: fixed	Short natural-language messages and machine-readable decisions. Deliver each active sender's message to all four neighbours next round. No chosen listening or repetition treatment. This mode needs implementation checks.
Model: planned comparison	Six named models are listed below: GPT-4.1 mini, Claude Haiku 4.5, Qwen3.8 27B, DeepSeek V4 Flash, Mistral Small 3.2, and Apertus 70B. Use one model for all agents in a game. Apertus access remains pending.
Language: planned comparison	Use the model-specific local language listed below: English, Chinese, French, or Standard German. Also compare every model in English and Vietnamese. Only English has been tested.
Source identity: experiment 2	The six labels below combine generic roles and a named figure in one experiment. Only ordinary-peer labels have been tested.
Network: experiment 3	Ring or connected random-regular graph; ten agents and four neighbours each in both.

The protocol compares six proposed models using shared English and Vietnamese conditions and additional model-specific languages. Appendix B gives the model roster, language assignments, and implementation details.

3.3 Shared procedure for all three experiments

1. Choose the model, language, disaster state, claim accuracy, and source position from a balanced scenario schedule. One controlled source receives the truth; all other agents start without that observation.

2. The source sends its assigned correct or false claim on rounds one and two. Keep that wording fixed within a language, apart from the identity label. Hold it active for those rounds, then stop its transmissions in every condition. Exclude it from recipient outcome rates. This controlled source is an explicit exception to ordinary agents' action rules.

3. Each other active agent reads its local history, sends a short message, and chooses stay or evacuate. It can answer or cite received messages. Deliver messages to all four neighbours on the next round. Evacuated agents stop sending.

4. End the game at a hidden terminal round drawn from the same four-to-six-round schedule. Score actions against the simulator's true state.

5. Repeat matched scenarios for the comparison conditions. Randomize execution order and use equal numbers of scenario seeds and repeated model runs per condition. Select the numerical sample size and message limit before evaluation, after compatibility checks.

The two-round source schedule is a fixed design choice, not a repetition treatment. Ordinary agents generate their messages live; only the initial source is controlled. In false-claim games there is no second informed source. These games test response to misleading information, not recovery after independent correction; paper C studies that separate question.

3.4 Experiment 1: language

Run each model in the languages listed above. Use the ordinary-peer source and ring network. Compare English with Vietnamese within every model, and English with the assigned local language where different. Translate meaning, rather than changing the facts or adding persuasion. Report language-following errors and translation-review status.

This tests the language of the whole game, including instructions and conversation. It does not isolate message language from instruction language. Cross-model comparisons in different local languages describe combined settings; they cannot identify a language effect on their own.

3.5 Experiment 2: claimed source identity

Use one combined identity experiment, with these six proposed labels translated consistently into every test language:

1. An ordinary teammate.
2. An emergency-management expert.
3. A senior Chinese government official.
4. Xi Jinping, President of China.
5. A Buddhist monk.
6. A Christian pastor.

Compare each authority role with the teammate. Compare the named political label with the generic political label as a more specific contrast. The latter changes name recognition and specificity of status; it does not perfectly isolate a name from rank. Monk and pastor are separate role conditions, not interchangeable religions. The Buddha and claims to speak for God remain possible follow-ups, outside this initial six-label matrix.

Show every selected label to every model, in each model's listed test languages. Do not reserve Xi Jinping for Chinese models. Keep message content, true evidence, graph, source position, and timing matched within each model–language comparison. The sender is not prompted to impersonate the named person. Recipients see a fictional attribution that grants no permissions.

For each model, compare the identity effect in English with its effect in Vietnamese and, where applicable, the local language. This tests whether language changes the response to an identity. It does not measure religious belief or political allegiance.

3.6 Experiment 3: network structure

Compare a ring, where each agent connects to the nearest two agents on each side, with a connected random-regular graph. Both have ten agents, four neighbours per agent, and twenty undirected edges. Use an ordinary-peer source and the same language conditions as above.

Generate several random graphs using recorded seeds. Rotate the source across positions and pair disaster state, accuracy, model, language, and terminal round. Preserve message budgets and delivery rules. Do not compare one lucky random graph with one source placement. Agents see local neighbours only, not a description of the global graph.

Measure whether the warning reaches more agents and whether their decisions improve. Changing network shape can change clustering and path lengths together. This experiment estimates their combined effect; it does not separately identify each graph property.

3.7 Outcomes, comparisons, and run counts

- **Missed danger:** Fraction of the nine non-source agents staying when disaster strikes.
- **False alarms:** Fraction evacuating when there is no disaster.
- **Information spread:** Who receives or relays the claim, how many rounds it takes, and what actions follow. Audit paraphrases; a cited ID alone does not prove agreement.
- **Other checks:** Payoff, action timing, invalid outputs, language errors, and refusals.

Use whole games as the units of analysis. Pair matched scenarios and report differences with uncertainty across games. Agents in the same game are not independent samples. Decide primary contrasts and how to account for multiple comparisons before evaluation. A message being received, cited, or followed by an action does not reveal private belief.

With all six models available there are sixteen model–language combinations. Experiment 1 needs sixteen ordinary-peer ring cells. Experiment 2 has ninety-six cells (six labels times sixteen combinations), including those same sixteen baselines. Experiment 3 has thirty-two cells (two networks times sixteen), also sharing the baselines. Thus there are **112 unique configuration cells**, before crossing the four disaster/accuracy cases and repeating scenarios. If each case has S scenario seeds and R repeated runs, the total is 112 times 4 times S times R games. Shared baselines must be counted once and their dependence retained in analysis. This is a planning calculation, not a launched batch or approved spending amount.

4. Results

4.1 Calibration status; proposed evaluation pending

A small calibration batch used `openai/gpt-4.1-mini`, English, ordinary-peer labels, and structured Safe/Danger signals. It completed 129 of 130 planned outcomes across A and C; one A run failed message-reference validation. The repository contains the full calibration report. These runs used the earlier repetition/listening design and the separate recovery study. They do not test the three experiments in this revised proposal. Keep those calibration artifacts unchanged.

Natural-language arguments, random-regular networks, identity labels, and the broader comparisons above remain planned. The current simulator must be updated and checked before this protocol can run. No results about Xi Jinping, the Buddha, God, languages other than English, or differences between model origins are claimed. The papers remain proposals. Before a larger evaluation, freeze prompts, model versions, message limits, sample sizes, stopping rules, and treatment order. Keep failed runs visible rather than silently dropping them.

5. Discussion and Limitations

The game is a simplified test of communication and action. A result on one ring network does not establish how all agent networks behave. Repeating the same model many times does not create many independent model families. Use whole games as the units of analysis, not individual agents, because agents influence each other. Use only simulated actions; external tools and hidden chain-of-thought are not needed.

The unpublished *Authority Across Languages* proposal informed the source-label and language questions. It is background, not evidence of experimental results. We make no claim that this is the first study of these topics.

6. Conclusion

This proposal defines comparisons to run, not findings from the revised evaluation. Its value must be assessed using the planned controlled experiments and reported failures as well as successful runs. No claim of proven safety improvement is made.

Code and Data

Project repository; scenario data in `data/evacuation/`; earlier calibration outcomes and analysis in `reports/2026-11-08-calibration/`. The existing runner and calibration artifacts do not implement or validate every proposed condition.

References

[1] C. I. Hovland and W. Weiss (1951). *The Influence of Source Credibility on Communication Effectiveness*. Public Opinion Quarterly, 15(4):635–650. https://doi.org/10.1086/266350.

[2] H. Shirado, F. W. Crawford, and N. A. Christakis (2020). *Collective communication and behaviour in response to uncertain 'Danger' in network experiments*. Proceedings of the Royal Society A, 476:20190685. https://doi.org/10.1098/rspa.2019.0685.

[3] S. Schachter (1951). *Deviation, rejection, and communication*. Journal of Abnormal and Social Psychology, 46:190–207. https://doi.org/10.1037/h0062326. Original experimental report also appears in *Theory and Experiment in Social Communication* (1950), https://cdn.isr.umich.edu/pubFiles/historicPublications/Theory_4867_.PDF.

[4] A. N. Tak et al. (2026). *From Process Loss to Assembly Bonus: Human-Grounded Diagnosis of Multi-Agent LLM Collaboration*. https://arxiv.org/html/2609.13261v1.

A. Limitations and Dual-Use / Ethical Considerations

The completed calibration used structured signals in English. It provides no estimate of how language, claimed identity, or network structure affects evacuation in the proposed experiments. Agents share a model within each game and act under artificial payoffs. Their responses cannot establish how speakers of a language or members of a religious or political community would respond to an emergency.

A finding that a false warning travels farther under a particular sender label or network structure could help someone impersonate an authority or target a group with fabricated alerts. In this protocol all attributions are fictional and remain inside the simulator. Report missed evacuations alongside unnecessary evacuations, disclose translation uncertainty, and present group-level outcomes. Avoid presenting an effective false message as a real quotation or publishing a ranking of communities by supposed susceptibility.

B. Model and Language Protocol

B.1 Named models for the planned comparison

We propose the following six-model roster for the three experiments. This is a list of specific test models, not a list of all models available in each region. Only GPT-4.1 mini has completed the calibration batch. The other five need a task-comprehension and output-format check before evaluation.

Table 3. Proposed model roster and access status.

Developer / location	Model, exact identifier, and status
OpenAI / US	GPT-4.1 mini: `openai/gpt-4.1-mini`. OpenRouter; calibration completed.
Anthropic / US	Claude Haiku 4.5: `anthropic/claude-haiku-4.5`. OpenRouter; planned.
Alibaba Qwen / China	Qwen3.8 27B: `qwen/qwen3.8-27b`. OpenRouter; planned.
DeepSeek / China	DeepSeek V4 Flash (0423): `deepseek/deepseek-v4-flash`. OpenRouter; planned.
Mistral AI / France	Mistral Small 3.2 24B Instruct: `mistralai/mistral-small-3.2-24b-instruct`. OpenRouter; planned.
Swiss AI Initiative / Switzerland	Apertus 70B Instruct (2509): `swiss-ai/Apertus-70B-Instruct-2509`. Hugging Face checkpoint; inference endpoint still to be arranged.

The first five IDs were checked against the OpenRouter model catalogue on November 8, 2026. Apertus is identified by its official model card; it was not found under the Swiss AI namespace in that catalogue. A catalogue entry is not a successful test call. We will record the resolved model version and serving provider for each run, and freeze reasoning settings as well as sampling settings. These models differ in age, size, and training, so this is not a capability-matched test of regions.

Use the same model for all ten agents within a game, then rerun the matched scenarios with the next model. If Apertus cannot be served within the pilot budget, report a five-model comparison and mark Apertus as pending. Do not silently substitute a second Mistral model or claim two independent European developers were tested.

B.2 Which language does each model use?

The main comparison uses a language associated with each developer's location, as listed below. We call this the *local-language condition*. Models do not have a human native language, and several locations have more than one language. For Switzerland, Standard German is our explicit study choice, not a claim that it is Apertus's only or best language.

Table 4. Language assignments for each model.

Model	Local-language condition	Shared baseline	Extra language
GPT-4.1 mini	English	English (same condition)	Vietnamese
Claude Haiku 4.5	English	English (same condition)	Vietnamese
Qwen3.8 27B	Simplified Chinese	English	Vietnamese
DeepSeek V4 Flash	Simplified Chinese	English	Vietnamese
Mistral Small 3.2	French	English	Vietnamese
Apertus 70B	Standard German	English	Vietnamese

What changes with language? Translate the task instructions, evidence, controlled claims, source labels, correction messages, and agent-facing conversation. Ask all agents to write their natural-language messages in that game's language. Keep machine-readable action keys and message IDs unchanged. Check translation meaning before evaluation, and record when an agent uses the wrong language or mixes languages. Do not silently count such runs as valid language-following examples.

Why retain English for every model? Comparing a Chinese model in Chinese with a French model in French changes two things at once. That comparison describes the combined model-and-language setup. Running both in English gives a shared baseline. Comparing the same model in English and its local-language condition tests language sensitivity within that model. English is counted only once for the US models.

Where does Vietnamese fit? Vietnamese is a separate shared-language test across all six models, including those whose developers are outside Vietnam. Reuse the same scenarios and treatments. This tests performance in Vietnamese; it does not make any model Vietnamese or measure Vietnamese culture. With all six models available, there are 16 distinct model–language combinations: six English, six Vietnamese, and four additional local-language conditions. This is a list of planned conditions, not a commitment to run every identity and treatment combination at once.

How to test identity fairly. Change the claimed sender label on the controlled message while keeping its words, evidence, timing, and network position the same. The label gives no extra permissions or true information. These are fictional attributions in the game, not real statements by those people or religious figures. Keep the correction sender fixed in C. Record refusal or role-play responses separately. Freeze exact labels and translations before running the comparison.

How to test models and languages fairly. Run the same scenarios and treatments for every selected model. Record the developer, exact model ID, access date, API host, and generation settings. A hosting provider is not the model's developer. Geographic labels describe the sample; they do not explain why a model behaves differently. Model ability, training, size, and language skill can all differ. We cannot infer national culture, political loyalty, or religious belief from these results.

LLM Usage Statement

An AI assistant helped investigate sources, design comparisons, write simulator code, and draft and edit this report. The recorded calibration outputs were checked by the repository's validators. Human authors must review the source interpretations, methods, and final text; independent human verification of every claim is not asserted.